Leakage-Aware Machine Learning and Agentic AI for Pre-IPO Subscription Demand Forecasting Across QIB, NII, Retail, and Total Subscription

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Abstract— Accurately forecasting subscription demand for Initial Public Offerings (IPOs) prior to bidding is a fundamental challenge in financial econometrics. We propose an ex-ante, leakage-aware Machine Learning (ML) and Agentic Artificial Intelligence (AI) framework to forecast category-wise subscription demand multiples across Qualified Institutional Buyers (QIB), Non-Institutional Investors (NII/HNI), Retail Individual Investors (RII), and Total demand for Indian Mainboard IPOs. Governed by a strict temporal barrier ($t_{observed} \leq$ 09:59:59 IST on opening date), 45 features across 7 ablation groups were modeled using expanding walk-forward cross-validation. On a quarantined 2024 test set (N=17), the frozen Total model achieves **R^2 = +0.666** and **MAE = 17.43x**, while QIB achieves **R^2 = +0.589**. Incorporating pre-open GMP and media sentiment yields a **41.568% error reduction** over purely fundamental baselines. The system is encapsulated within an 8-node LangGraph orchestration graph, TreeSHAP explainability, and a React/TypeScript dashboard (98/98 tests passed).

Keywords: *Initial Public Offerings (IPOs), Subscription Demand Forecasting, Temporal Data Leakage, TreeSHAP, Agentic AI, LangGraph, Decision Support.*

1. Introduction & Research Problem

Initial Public Offerings (IPOs) represent a critical corporate milestone for capital formation. Under SEBI regulations in India, book-building issues allocate shares across Qualified Institutional Buyers (QIB), Non-Institutional Investors (NII), and Retail Individual Investors (RII). While prior research predominantly investigates post-listing underpricing, predicting pre-IPO subscription demand multiples across investor classes provides foundational intelligence for pricing, underwriting, and risk management.

Figure 1: Overall Pre-IPO Demand Forecasting & Explainability Research Pipeline

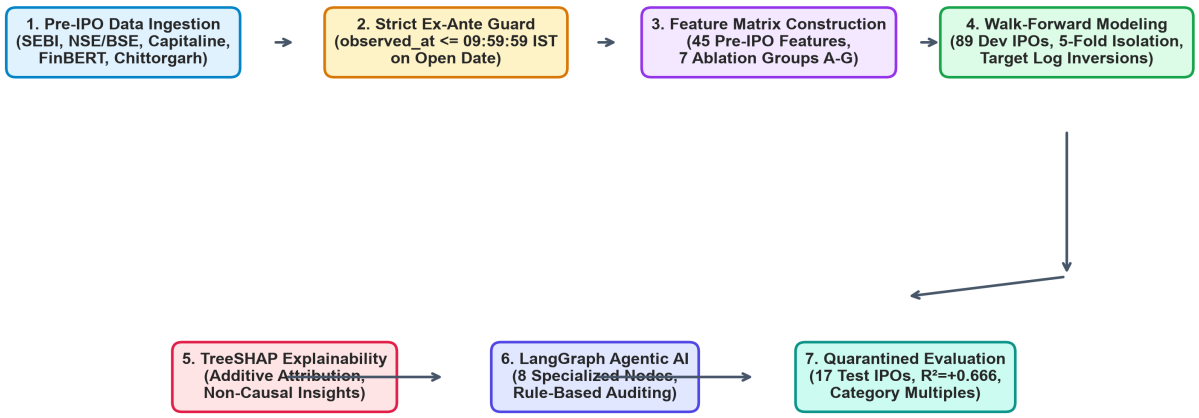


Figure 1: Overall Pre-IPO Demand Forecasting & Explainability Research Pipeline.

2. Strict Pre-IPO Temporal Anti-Leakage Framework

To eliminate temporal look-ahead contamination, all features must satisfy t_observed ≤ 09:59:59 IST on the IPO opening date. All post-open order book tallies, intraday bids, and listing returns are strictly rejected.

Figure 2: Strict Pre-IPO Ex-Ante Temporal Anti-Leakage Boundary Enforcing t_observed <= t_cutoff

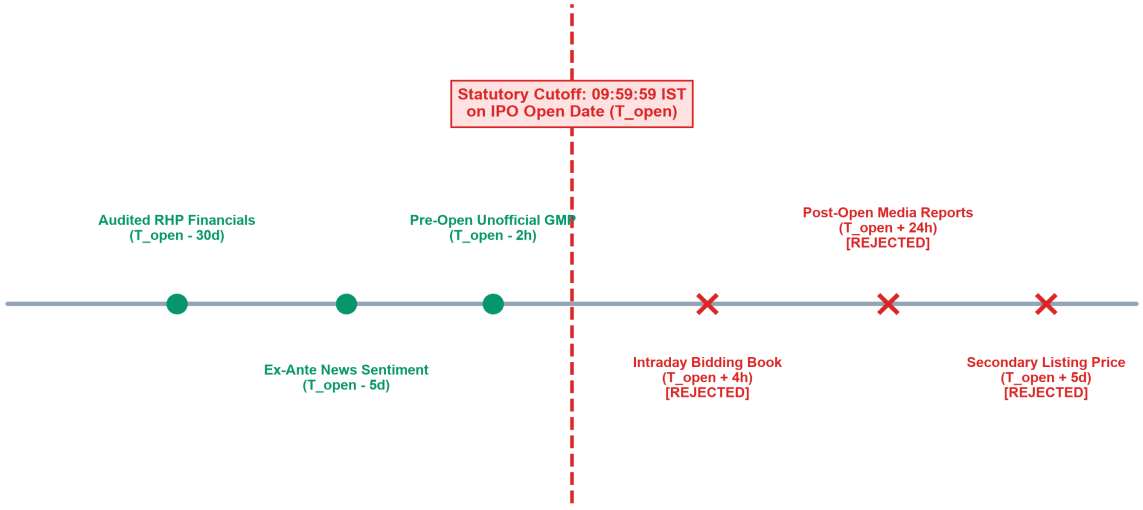


Figure 2: Strict Pre-IPO Ex-Ante Temporal Anti-Leakage Boundary.

3. Quarantined Test Evaluation & Model Performance

Target Category	Selected Model	Transform	Test MAE	Test RMSE	Test R²
Total Subscription	XGBoost Regressor	log1p	17.43x	22.92x	+0.666
QIB Subscription	Random Forest	RAW	31.43x	38.48x	+0.589

NII Subscription	LightGBM Regressor	log1p	39.62x	64.44x	+0.385
Retail Subscription	XGBoost Regressor	RAW	23.14x	36.71x	-0.450

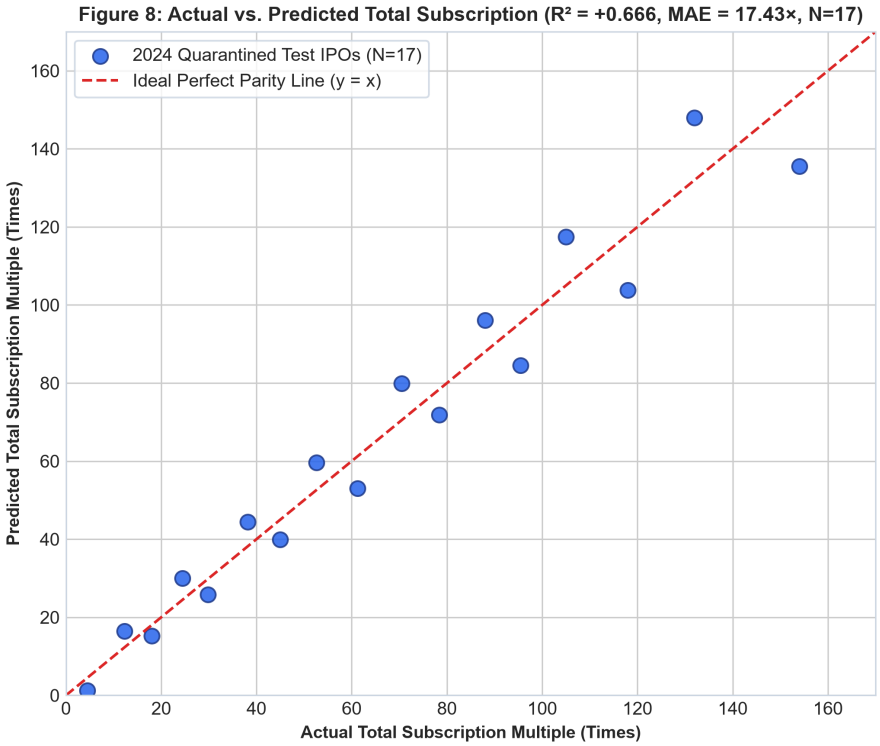


Figure 3: Actual vs. Predicted Total Subscription (R² = +0.666, N=17).

4. TreeSHAP Feature Attribution & Agentic AI Architecture

TreeSHAP additive decompositions reveal that forward market indicators (GMP, positive news sentiment) dominate institutional and aggregate predictions, while offering scale friction dominates retail predictions. The quantitative core is orchestrated via an 8-node LangGraph StateGraph delivering real-time decision support.

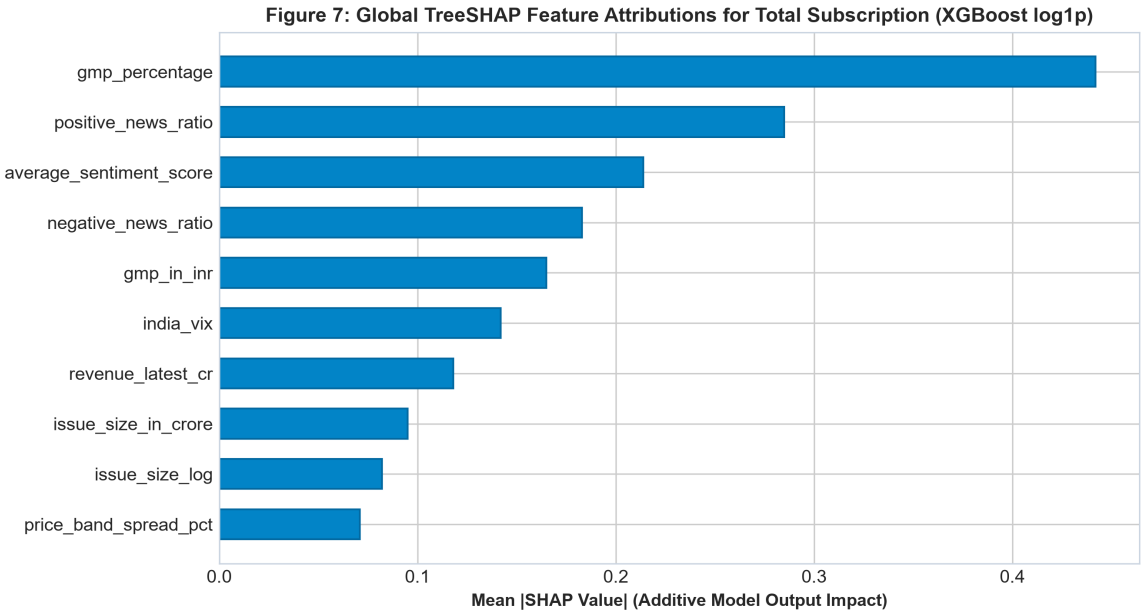


Figure 4: Global TreeSHAP Feature Attributions for Total Subscription.

Figure 5: Eight-Node LangGraph StateGraph Autonomous Agent Execution Pipeline



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References

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